HW 2 Jacob Callebs

① $x^3 - 7x^2 + 14x - 5 = 0$ for $e < 0.01$

A.) Bisection $x_0 = 0$ $x_1 = 1$

$x_2 = \frac{x_0 + x_1}{2}$ If $f(x_2) < 0$, swap x_0 else, swap x_1

i	x_0	x_1	x_2	$f(x_2)$	$e = \lvert x^{new} - x^{old} \rvert$
1	0	1	0.5	0.375	0.5
2	0	0.5	0.25	-1.921	0.25
3	0.25	0.5	0.375	-0.682	0.125
4	0.375	0.5	0.438	-0.127	0.063
5	0.438	0.5	0.469	0.129	0.031
6	0.438	0.469	0.454	0.007	0.015
7	0.438	0.454	0.446	-0.06	0.008

Note: Values are rounded to 3 decimals and used in that form for each calculation. I say $i=0$ is guess x_0 and x_1

B.) Newton - Raphson $x_0 = 0$

$f(x) = x^3 - 7x^2 + 14x - 5$

$f'(x) = 3x^2 - 14x + 14$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

i	x_n	x_{n+1}	$f(x_{n+1})$	$e = \lvert x^{new} - x^{old} \rvert$
1	0	0.357	-0.849	0.357
2	0.357	0.447	-0.051	0.09
3	0.447	0.453	-0.0015	0.006

I say $i=0$ is guess x_0

Much faster than Bisection! Half the iterations

② A.) Bisect $i = 28$ True $R = 0.453181723$

N-R $i = 5$

B.) Keep $\varepsilon = 1e-8$

Bisection

$x_L = 0.2$ $x_R = 0.6$ $\Rightarrow i = 27$

$x_L = -2$ $x_R = 5$ $\Rightarrow i = 31$

$x_L = -2$ $x_R = 15$ $\Rightarrow i = 32$

$x_L = -34$ $x_R = 15$ $\Rightarrow i = 34$

Stable

N-R

$x_0 = -15$ $\Rightarrow i = 11$

$x_0 = 0.3$ $\Rightarrow i = 5$

$x_0 = 2$ $\Rightarrow i = 44$

$x_0 = 5$ $\Rightarrow i = 36$

$x_0 = 15$ $\Rightarrow i = 25$

C.) $f'(x) = 3x^2 - 14x + 14$

$$x = \frac{14 \pm \sqrt{14^2 - 4(3)(14)}}{6}$$

$$x = \frac{14 \pm \sqrt{28}}{6}$$

Here, slope $= 0$